

Linear Regression Model Report (California Housing Dataset)

1. Introduction

The objective of this project is to build and evaluate a machine learning model that predicts house prices using the California Housing dataset. The project demonstrates the complete machine learning workflow including data loading, exploration, preprocessing, model training, evaluation, and visualization.

2. Dataset Information

The California Housing dataset is provided by scikit-learn and contains information about housing districts in California. The dataset includes numerical features such as median income, housing age, average rooms, average bedrooms, population, occupancy, latitude, and longitude.

Target Variable: MedHouseVal (Median House Value)

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset structure, feature distributions, correlations, and missing values.

- Checked dataset shape, column names, and statistical summary.
- Verified that the dataset contains no missing values.
- Generated histograms to visualize feature distributions.
- Used a heatmap to analyze correlations between features and the target variable.
- Observed that Median Income showed a strong positive correlation with house prices.

4. Model Used

A Linear Regression model from scikit-learn was used for prediction. The dataset was divided into training and testing sets using `train_test_split` with an 80:20 ratio.

5. Evaluation Metrics

The following evaluation metrics were used to measure model performance:

- MAE (Mean Absolute Error): Measures average prediction error.
- RMSE (Root Mean Squared Error): Measures error magnitude with higher penalty for large errors.
- R^2 Score: Indicates how well the model explains the variance in house prices.

Typical R^2 values for this Linear Regression model on the California Housing dataset are around 0.57 to 0.60.

6. Conclusion

The Linear Regression model successfully learned relationships between housing features and house prices. The project demonstrates the complete machine learning pipeline including data preprocessing, training, evaluation, and visualization.

7. Improvements:

Apply feature scaling and preprocessing techniques.

- Use advanced regression models such as Random Forest or XGBoost.
- Perform hyperparameter tuning.
- Deploy the model using Streamlit or Flask.
- Tools and Libraries Used
 - Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - Jupyter Notebook